

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

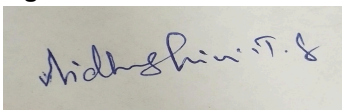
Total Marks:40

Code of Conduct:

I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough → Possible Flu
- II. If a patient has Fever AND Rash → Possible Measles
- III. If a patient has Cough AND Breathlessness → Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

		analyze the case deeply			
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 – Smart Medical Diagnosis System :

(a) Propositional Logic Representation

Let the propositions be:

- F = Patient has Fever
- C = Patient has Cough
- B = Patient has Breathlessness
- Flu = Possible Flu
- M = Possible Measles
- P = Possible Pneumonia

The given rules can be represented as:

Rule 1: $F \wedge C \rightarrow \text{Flu}$

Rule 2: $F \wedge \text{Rash} \rightarrow M$

Rule 3: $C \wedge B \rightarrow P$

Facts about Patient A:

$F = \text{True}, C = \text{True}, B = \text{True}$

So, the knowledge base contains the three rules and these three facts.

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules whose conditions are satisfied.

Step 1: Known facts:

Fever(A)=True Cough(A)=True Breathlessness(A)=True

Step 2: Apply Rule 1

Rule: $\text{Fever} \wedge \text{Cough} \rightarrow \text{Flu}$

Both Fever and Cough are true.

Therefore: PossibleFlu(A)=True

Step 3: Apply Rule 3

Rule: $\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Pneumonia}$

Both Cough and Breathlessness are true.

Therefore: PossiblePneumonia(A)=True

Step 4: Rule 2

Rule 2 requires Fever and Rash. Although Fever is true, Rash is not given as a fact.

Therefore, Measles cannot be derived.

(c) Backward Chaining for Pneumonia

Goal: Pneumonia(A)

To prove this goal, search for a rule whose conclusion is Pneumonia.

The relevant rule is:

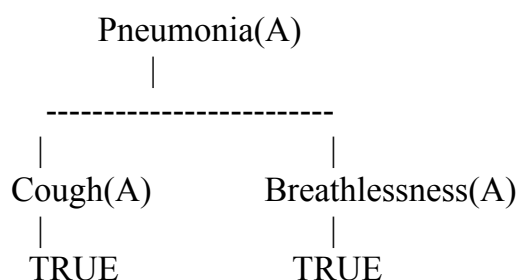
$\text{Cough}(x) \wedge \text{Breathlessness}(x) \rightarrow \text{Pneumonia}(x)$

Therefore, the goal is reduced into two sub-goals:

Cough(A) and Breathlessness(A)

Both are already given as facts.

Goal Reduction Tree:



Since both sub-goals are true, the original goal is proved.

Case Study 2 – Autonomous Traffic Management Agent

The knowledge base uses First-Order Logic (FOL) with variables and predicates such as Vehicle, EmergencyType, ClearPath, GreenSignal, Behind and Proceed.

(a) Unification

Rule 1: $\forall x(\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x))$

Given facts:

Vehicle(Ambulance) EmergencyType(Ambulance)

We need to match the rule variable x with the constant Ambulance.

Step 1: Match:

Vehicle(x) with Vehicle(Ambulance)

Therefore: $x/\text{Ambulance}$

Step 2: Check the second condition:

EmergencyType(x)

After substitution: EmergencyType(Ambulance)

This matches the given fact.

Therefore, the substitution set is:

$\theta = \{x/\text{Ambulance}\}$

After applying the substitution:

ClearPath(Ambulance) is derived.

(b) Resolution Proof for Proceed(CarA)

Step 1: Convert relevant rules into CNF

Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Using $A \wedge B \rightarrow C$:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2: $\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This gives:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ and $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3: $\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Step 2: Use the facts

Given: $\text{Vehicle}(\text{Ambulance}) \text{ EmergencyType}(\text{Ambulance})$

From Rule 1: $\text{ClearPath}(\text{Ambulance})$

Using Rule 2: $\text{GreenSignal}(\text{Ambulance})$

Other given facts:

$\text{Vehicle}(\text{CarA}) \text{ Behind}(\text{CarA}, \text{Ambulance})$

Step 3: Apply Rule 3

Rule 3 requires:

- $\text{Vehicle}(\text{CarA})$
- $\text{Behind}(\text{CarA}, \text{Ambulance})$
- $\text{GreenSignal}(\text{Ambulance})$

All three are true.

Therefore:

$\text{Proceed}(\text{CarA})$

Resolution chain

$\text{Vehicle}(\text{Ambulance})$

+

$\text{EmergencyType}(\text{Ambulance})$

↓

$\text{ClearPath}(\text{Ambulance})$

↓

$\text{GreenSignal}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

$\text{GreenSignal}(\text{Ambulance})$

↓

$\text{Proceed}(\text{CarA})$

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle more than one vehicle because the rules use the variable x. However, the given facts contain only one emergency vehicle, Ambulance.

For two simultaneous emergency vehicles, we need additional facts such as:

Vehicle(Ambulance2) EmergencyType(Ambulance2)

Then Rule 1 can derive: ClearPath(Ambulance2)

and Rule 2 can derive: GreenSignal(Ambulance2)

We may also need facts describing their positions, for example:

Behind(CarB,Ambulance2)

Therefore, the basic FOL rules are reusable, but additional facts about the second emergency vehicle and its position are required.

Case Study 3 – Agricultural Expert Reasoning System

The propositional knowledge base contains three rules and two initial facts.

(a) Conversion into CNF

P1:

SoilDry $\rightarrow$ IrrigationNeeded

Step 1 – Eliminate implication:

$\neg$ SoilDry $\vee$ IrrigationNeeded

This is already in CNF.

Therefore: $\neg$ SoilDry $\vee$ IrrigationNeeded

P2:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Step 1 – Eliminate implication:

$\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod

Step 2 – Move negation inward:

($\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat) $\vee$ ApplyDripMethod

Therefore: $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

P3:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Eliminating implication:

$\neg(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Move negation inward:

$\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$

Therefore: $\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$

Facts : SoilDry CropWheat

So the complete CNF contains these five clauses.

(b) Resolution Refutation for ApplyDripMethod

Step 1: Goal

ApplyDripMethod

For resolution refutation, negate the goal: $\neg\text{ApplyDripMethod}$

Add this to the KB.

Step 2: Use SoilDry

From P1:

$\neg\text{SoilDry} \vee \text{IrrigationNeeded}$

Given: SoilDry

Resolution gives: IrrigationNeeded

Step 3: Apply P2

P2: $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

We have: IrrigationNeeded and CropWheat

Therefore: ApplyDripMethod

Step 4: Contradiction

We initially added: $\neg\text{ApplyDripMethod}$

ApplyDripMethod:

Resolving these two gives: The empty clause means a contradiction has been obtained.

Resolution Proof

Negate the goal: $\neg\text{ApplyDripMethod}$

Given clauses:

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: SoilDry ,

C4: CropWheat

Step 1: $C1 + C3 \Rightarrow \text{IrrigationNeeded}$

Step 2: $C2 + \text{IrrigationNeeded} \Rightarrow \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3: $(\neg \text{CropWheat} \vee \text{ApplyDripMethod}) + \text{CropWheat} \Rightarrow \text{ApplyDripMethod}$

Step 4: $\text{ApplyDripMethod} + \neg \text{ApplyDripMethod} \Rightarrow \{\}$

Therefore: ApplyDripMethod is proved

(c) Removing SoilDry

Initially: SoilDry was used to derive: IrrigationNeeded which then helped derive:

ApplyDripMethod

If SoilDry is removed, P1 cannot be activated.

Therefore: IrrigationNeeded cannot be derived from the remaining facts.

As a result, P2 cannot be used to conclude:

ApplyDripMethod

Now consider P3:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Although:

$\text{CropWheat} = \text{True}$

we do not have:

$\neg \text{ApplyDripMethod}$

Therefore, CropAtRisk also cannot be logically concluded.

Case Study 4 – Wumpus World Knowledge Agent

The given percepts are Stench at [1,2], Breeze at [1,1], and Glitter at [2,2]. The rules relate these percepts to possible Wumpus, Pit, Gold and Safe locations.

(a) Belief State and Modus Ponens

Given percepts:

1. Stench[1,2]
2. Breeze[1,1]
3. Glitter[2,2]

Step 1: Stench

Rule:

Stench[1,2] → Wumpus adjacent to [1,2]

Since Stench[1,2] is true: Wumpus is adjacent to [1,2]

Step 2: Breeze

Rule:

Breeze[1,1] → Pit adjacent to [1,1]

Since Breeze[1,1] is true: Pit is adjacent to [1,1]

Step 3: Glitter

Rule: Glitter[x,y] → Gold[x,y]

For [2,2]: Glitter[2,2] → Gold[2,2]

Therefore: Gold[2,2]

Belief state

The agent knows:

- There is a possible Wumpus adjacent to [1,2].
- There is a possible Pit adjacent to [1,1].
- Gold is present at [2,2].
- No Safe[x,y] conclusion can be made yet because the required negative Wumpus and Pit facts are not given.

(b) Forward Chaining for Wumpus and Pit

Wumpus reasoning

Given: Stench[1,2]

Apply Rule 1: Wumpus adjacent to [1,2]

In a normal grid with four-directional adjacency, the possible neighboring cells are:

[1,1],[1,3],[2,2]

Therefore, the Wumpus may be in one of these cells based on the information provided.

Pit reasoning

Given: Breeze[1,1]

Apply Rule 2: Pit adjacent to [1,1]

The possible neighboring cells are:

[1,2],[2,1]

Therefore, the Pit may be in [1,2] or [2,1].

Stench[1,2]

↓

Wumpus adjacent to [1,2]

↓

Possible: [1,1], [1,3], [2,2]

Breeze[1,1]

↓

Pit adjacent to [1,1]

↓

Possible: [1,2], [2,1]

(c) Safety of the Path

Proposed path: [1,1]→[1,2]→[2,2]

Cell [1,1]

The agent perceives: Breeze[1,1]

Cell [1,2]

The agent perceives: Stench[1,2]

Also, [1,2] itself is a possible Pit location from the Breeze at [1,1].

The agent perceives: Glitter[2,2]

Therefore: Gold[2,2].